Aritra Biswas

Senior Manager, AI Engineering

pandalearnstocode@gmail.com | +91-9051237918 | Bengaluru, India | aritro.in | pandalearnstocode
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Profile

AI engineering and platform leader with 9+ years across data science, machine learning, and production software, including 3+ years scaling multi-region teams to 30+ engineers. Set quarterly AI strategy, own a $3M engineering budget, and deliver reliable enterprise platforms for 10K+ users at 99.9% uptime. Outcomes include 28% lower LLM cost, approximately $850K productivity value, and $12M measured revenue.

Experience

Senior Manager – AI Engineering | AB InBev, Bengaluru ..... Oct 2023 – Present

- Scaled the organization from 8 to 30+ across India, Mexico, and Brazil; cut hiring time from 90 to 45 days and sustained 88% retention. Lead quarterly AI strategy, cross-functional executive steering, build-versus-buy/ROI reviews, a $3M headcount/cloud/vendor budget, and architecture/NFR standards. Microsoft and Databricks partnerships save $120K annually.
- Architected and set platform strategy for an AI Gateway governing $700K in annual LLM spend and serving 10K+ users at 99.9% availability. Intelligent routing, semantic caching/fallbacks, prompt optimization, and evaluation reduced cost 28%, latency 40%, and improved response quality 20%; established product FinOps, LLMOps, and operational excellence through LangFuse/Datadog observability.
- Delivered a five-product AI portfolio: Cheers (executive productivity; MVP in one week, production in two months), CPG Intel (competitor monitoring), Social Listening (sentiment/anomaly detection), Insights Copilot (5K+ TAU, 1K+ MAU, 300+ concurrent users, under $0.02/query), and AlgoVault (B2B recommendations in 13 countries, 50K+ daily predictions, $12M measured revenue).

Senior Machine Learning Engineer | AB InBev, Bengaluru ..... Sep 2021 – Sep 2023

- Secured $1.2M after delivering 18% cross-sell uplift; scaled hybrid recommendation models across 13 countries and 50K+ daily predictions. Ray/Spark training processed 5M+ transactions; Redis-backed inference reached sub-100ms P95. Built evaluation coverage for Precision@K, Recall@K, NDCG, and MAP.
- Cut Bayesian MMM runtime 85% and hyperparameter optimization from 3 days to 4 hours; supported $500M+ marketing investment, 8–12% MROI improvement, and 6–9% gains on a $180M allocator.
- Promoted to Senior Manager for technical impact; led a five-person data science team and MIT BudLab research collaboration. Productionized ML on AKS/FastAPI/Docker/GitHub Actions CI/CD (3x throughput; weeks to days); migrated 15K+ R lines to Python, introduced feature store/MLflow, champion-challenger releases, drift detection and rollback, and doubled delivery velocity at 93%+ model accuracy.

Data Scientist | AB InBev, Bengaluru ..... May 2020 – Aug 2021

- Built enterprise causal marketing measurement for Brazil, Mexico, and Colombia using Difference-in-Differences, Synthetic Control, and Propensity Score Matching. LLVM-optimized DTW ran 50x faster; analysis of 5M+ transactions and $85M media spend identified $8M in inefficiency and improved ROAS from 2.1x to 2.8x.

Data Scientist | Nielsen – Bengaluru ..... Mar 2017 – May 2020

- Built Rapid MMM SaaS for in-store lift and budget allocation, deployed and trained across seven accounts. StoryBoarder supported $95M revenue and 120+ engagements. Led a three-person team, integrated 50+ sources, reduced data-quality incidents 75%, and supported 50% engagement growth without added headcount.

Technical Skills

Leadership: Engineering management, organization scaling, hiring/retention, coaching, cross-functional/executive alignment, AI strategy, portfolio/ROI, build-versus-buy, budget ownership, architecture governance
AI/ML: GenAI/LLMs, LLMOps, evaluation, FinOps, observability, recommendation systems, MMM, causal inference, distributed training
Platforms: Azure AI Foundry/ML, Databricks, AKS/Kubernetes, APIM, MLflow, KEDA, Redis, LangChain/LangGraph, LiteLLM, MCP/A2A, vector DBs
Engineering: Python, FastAPI, SQL, Spark/Ray, REST/microservices, Docker, GitHub Actions, Datadog/Langfuse, Neo4j, Elasticsearch, LLVM, CuPy, Numba, Cython
AI-assisted development: Claude, Codex, GitHub Copilot, Codespaces, Dev Containers

Education

M.Sc., Statistics | University of Delhi 2017
B.Sc., Statistics | Presidency College, University of Calcutta 2015